

SKILL(WEEK-3) DSEDBD SEC-10

2520090046

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1. Create Tables for Joins

```
3 CREATE TABLE class (  
4   id INT,  
5   name VARCHAR(30)  
6 );  
7 CREATE TABLE class_info (  
8   id INT,  
9   address VARCHAR(30)  
10 );  
11
```

Output			
Action Output			
#	Time	Action	Message
28	12:38:00	CREATE TABLE book1 (book_id INT AUTO_INCREMENT PRIMARY KEY, title VARCHAR(255) NOT NULL...	Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds to your MySQL ser...
29	13:31:27	USE school_db	0 row(s) affected
30	13:31:33	CREATE DATABASE IF NOT EXISTS school_db	1 row(s) affected, 1 warning(s): 1007 Can't create database 'school_db'; database exists
31	13:31:51	USE school_db	0 row(s) affected
32	13:32:23	CREATE TABLE class (id INT, name VARCHAR(30))	0 row(s) affected
33	13:32:28	CREATE TABLE class_info (id INT, address VARCHAR(30))	0 row(s) affected

2. Insert Data for Cross Join

```
11 INSERT INTO class VALUES  
12 (1, 'abhi'),  
13 (2, 'adam'),  
14 (4, 'alex');  
15 INSERT INTO class_info VALUES  
16 (1, 'DELHI'),  
17 (2, 'MUMBAI'),  
18 (3, 'CHENNAI');
```

Output			
Action Output			
#	Time	Action	Message
30	13:31:33	CREATE DATABASE IF NOT EXISTS school_db	1 row(s) affected, 1 warning(s): 1007 Can't create database 'school_db'; database exists
31	13:31:51	USE school_db	0 row(s) affected
32	13:32:23	CREATE TABLE class (id INT, name VARCHAR(30))	0 row(s) affected
33	13:32:28	CREATE TABLE class_info (id INT, address VARCHAR(30))	0 row(s) affected
34	13:33:46	INSERT INTO class_info VALUES (1, 'DELHI'), (2, 'MUMBAI'), (3, 'CHENNAI')	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
35	13:33:50	INSERT INTO class_info VALUES (1, 'DELHI'), (2, 'MUMBAI'), (3, 'CHENNAI')	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0

3. Cross Join Query

```
19 SELECT *  
20 FROM class  
21 CROSS JOIN class_info;
```

Result Grid			
#	id	name	id
1	1	abhi	1
2	2	adam	1
3	4	alex	1
4	1	abhi	2
5	2	adam	2
6	4	alex	2
7	1	abhi	3
8	2	adam	3
9	4	alex	3

Output			
Action Output			
#	Time	Action	Message
31	13:31:51	USE school_db	0 row(s) affected
32	13:32:23	CREATE TABLE class (id INT, name VARCHAR(30))	0 row(s) affected
33	13:32:28	CREATE TABLE class_info (id INT, address VARCHAR(30))	0 row(s) affected
34	13:33:46	INSERT INTO class_info VALUES (1, 'DELHI'), (2, 'MUMBAI'), (3, 'CHENNAI')	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
35	13:33:50	INSERT INTO class_info VALUES (1, 'DELHI'), (2, 'MUMBAI'), (3, 'CHENNAI')	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
36	13:34:28	SELECT * FROM class CROSS JOIN class_info LIMIT 0, 1000	0 row(s) returned

4. Reset Data for Main Joins

```
41 TRUNCATE TABLE class_info;
22 • TRUNCATE TABLE class;
23 • TRUNCATE TABLE class_info;
24 • INSERT INTO class VALUES
25 (1, 'abhi'),
26 (2, 'adam'),
27 (3, 'alex'),
28 (4, 'anu'),
29 (5, 'ashish');
38 • INSERT INTO class_info VALUES
31 (1, 'DELHI'),
32 (2, 'MUMBAI'),
33 (3, 'CHENNAI'),
34 (7, 'NOIDA'),
35 (8, 'PANIPAT');
36
```

Output

#	Time	Action	Message	Duration / Fetch
36	13:34:28	SELECT * FROM class CROSS JOIN class_info LIMIT 0, 1000	0 row(s) returned	0.000 sec / 0.000 sec
37	13:36:27	TRUNCATE TABLE class	0 row(s) affected	0.031 sec
38	13:36:30	TRUNCATE TABLE class_info	0 row(s) affected	0.016 sec
39	13:36:33	TRUNCATE TABLE class_info	0 row(s) affected	0.016 sec
40	13:36:45	INSERT INTO class VALUES (1, 'abhi'), (2, 'adam'), (3, 'alex'), (4, 'anu'), (5, 'ashish')	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.000 sec
41	13:36:50	INSERT INTO class_info VALUES (1, 'DELHI'), (2, 'MUMBAI'), (3, 'CHENNAI'), (7, 'NOIDA'), (8, 'PANIPAT')	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.016 sec

5. Inner Join Queries

```
35 (8, 'PANIPAT');
36 • SELECT *
37 FROM class
38 INNER JOIN class_info
39 ON class.id = class_info.id;
40 • SELECT class.name, class_info.address
41 FROM class
42 INNER JOIN class_info
43 ON class.id = class_info.id;
```

Result Grid

	name	address
▶	abhi	DELHI
	adam	MUMBAI
	alex	CHENNAI

Result 4 x

Output

#	Time	Action	Message	Duration / Fetch
39	13:36:33	TRUNCATE TABLE class_info	0 row(s) affected	0.016 sec
40	13:36:45	INSERT INTO class VALUES (1, 'abhi'), (2, 'adam'), (3, 'alex'), (4, 'anu'), (5, 'ashish')	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.000 sec
41	13:36:50	INSERT INTO class_info VALUES (1, 'DELHI'), (2, 'MUMBAI'), (3, 'CHENNAI'), (7, 'NOIDA'), (8, 'PANIPAT')	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.016 sec
42	13:37:56	SELECT class.name, class_info.address FROM class INNER JOIN class_info ON class.id = class_info.id LIMIT ...	3 row(s) returned	0.000 sec / 0.000 sec
43	13:38:23	SELECT class.name, class_info.address FROM class INNER JOIN class_info ON class.id = class_info.id LIMIT ...	3 row(s) returned	0.000 sec / 0.000 sec
44	13:38:31	SELECT class.name, class_info.address FROM class INNER JOIN class_info ON class.id = class_info.id LIMIT ...	3 row(s) returned	0.000 sec / 0.000 sec

6. Natural Join Query

```
43 ON class.id = class_info.id;
44 • SELECT *
45 FROM class
46 NATURAL JOIN class_info;
```

Result Grid

	id	name	address
▶	1	abhi	DELHI
	2	adam	MUMBAI
	3	alex	CHENNAI

Result 5 x

Output

#	Time	Action	Message	Duration / Fetch
40	13:36:45	INSERT INTO class VALUES (1, 'abhi'), (2, 'adam'), (3, 'alex'), (4, 'anu'), (5, 'ashish')	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.000 sec
41	13:36:50	INSERT INTO class_info VALUES (1, 'DELHI'), (2, 'MUMBAI'), (3, 'CHENNAI'), (7, 'NOIDA'), (8, 'PANIPAT')	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.016 sec
42	13:37:56	SELECT class.name, class_info.address FROM class INNER JOIN class_info ON class.id = class_info.id LIMIT ...	3 row(s) returned	0.000 sec / 0.000 sec
43	13:38:23	SELECT class.name, class_info.address FROM class INNER JOIN class_info ON class.id = class_info.id LIMIT ...	3 row(s) returned	0.000 sec / 0.000 sec
44	13:38:31	SELECT class.name, class_info.address FROM class INNER JOIN class_info ON class.id = class_info.id LIMIT ...	3 row(s) returned	0.000 sec / 0.000 sec
45	13:39:08	SELECT * FROM class NATURAL JOIN class_info LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec

7. Left Outer Join Queries

```

47 SELECT *
48 FROM class
49 LEFT OUTER JOIN class_info
50 ON class.id = class_info.id;
51 SELECT *
52 FROM class
53 LEFT JOIN class_info
54 ON class.id = class_info.id
55 WHERE class_info.id IS NULL;

```

id	name	id	address
4	anu	NULL	NULL
5	ashish	NULL	NULL

Result 10 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
45	13:39:08	SELECT * FROM class NATURAL JOIN class_info LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
46	13:39:42	SELECT * FROM class NATURAL JOIN class_info LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
47	13:39:49	SELECT * FROM class LEFT OUTER JOIN class_info ON class.id = class_info.id LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec
48	13:39:56	SELECT * FROM class LEFT OUTER JOIN class_info ON class.id = class_info.id LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec
49	13:40:02	SELECT * FROM class LEFT JOIN class_info ON class.id = class_info.id WHERE class_info.id IS NULL LIMIT 0, 1000	2 row(s) returned	0.000 sec / 0.000 sec
50	13:40:08	SELECT * FROM class LEFT JOIN class_info ON class.id = class_info.id WHERE class_info.id IS NULL LIMIT 0, 1000	2 row(s) returned	0.000 sec / 0.000 sec

8. Right Outer Join Queries

```

56 SELECT *
57 FROM class
58 RIGHT OUTER JOIN class_info
59 ON class.id = class_info.id;
60 SELECT *
61 FROM class
62 RIGHT JOIN class_info
63 ON class.id = class_info.id
64 WHERE class.id IS NULL;

```

id	name	id	address
7	Noida	NULL	NULL
8	PANIPAT	NULL	NULL

Result 11 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
46	13:39:42	SELECT * FROM class NATURAL JOIN class_info LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
47	13:39:49	SELECT * FROM class LEFT OUTER JOIN class_info ON class.id = class_info.id LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec
48	13:39:56	SELECT * FROM class LEFT OUTER JOIN class_info ON class.id = class_info.id LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec
49	13:40:02	SELECT * FROM class LEFT JOIN class_info ON class.id = class_info.id WHERE class_info.id IS NULL LIMIT 0, 1000	2 row(s) returned	0.000 sec / 0.000 sec
50	13:40:08	SELECT * FROM class LEFT JOIN class_info ON class.id = class_info.id WHERE class_info.id IS NULL LIMIT 0, 1000	2 row(s) returned	0.000 sec / 0.000 sec
51	13:40:48	SELECT * FROM class RIGHT JOIN class_info ON class.id = class_info.id WHERE class.id IS NULL LIMIT 0, 1000	2 row(s) returned	0.000 sec / 0.000 sec

9. Full Outer Join Queries

```

73 SELECT *
74 FROM class
75 LEFT JOIN class_info ON class.id = class_info.id
76 WHERE class_info.id IS NULL
77 UNION
78 SELECT *
79 FROM class
80 RIGHT JOIN class_info ON class.id = class_info.id
81 WHERE class.id IS NULL;

```

id	name	id	address
4	anu	NULL	NULL
5	ashish	NULL	NULL
7	Noida	NULL	NULL
8	PANIPAT	NULL	NULL

Result 12 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
47	13:39:49	SELECT * FROM class LEFT OUTER JOIN class_info ON class.id = class_info.id LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec
48	13:39:56	SELECT * FROM class LEFT OUTER JOIN class_info ON class.id = class_info.id LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec
49	13:40:02	SELECT * FROM class LEFT JOIN class_info ON class.id = class_info.id WHERE class_info.id IS NULL LIMIT 0, 1000	2 row(s) returned	0.000 sec / 0.000 sec
50	13:40:08	SELECT * FROM class LEFT JOIN class_info ON class.id = class_info.id WHERE class_info.id IS NULL LIMIT 0, 1000	2 row(s) returned	0.000 sec / 0.000 sec
51	13:40:48	SELECT * FROM class RIGHT JOIN class_info ON class.id = class_info.id WHERE class.id IS NULL LIMIT 0, 1000	2 row(s) returned	0.000 sec / 0.000 sec
52	13:41:27	SELECT * FROM class LEFT JOIN class_info ON class.id = class_info.id WHERE class_info.id IS NULL UNION SELECT * FROM class RIGHT JOIN class_info ON class.id = class_info.id WHERE class.id IS NULL LIMIT 0, 1000	4 row(s) returned	0.000 sec / 0.000 sec

10. Setup Tables for Set Operations

```

82 • CREATE TABLE first_table (
83   id INT,
84   name VARCHAR(30)
85 );
86
87 • CREATE TABLE second_table (
88   id INT,
89   name VARCHAR(30)
90 );
91
92 • INSERT INTO first_table VALUES
93   (1, 'abhi'),
94   (2, 'adam');
95
96 • INSERT INTO second_table VALUES
97   (2, 'adam'),
98   (3, 'chester');

```

Output

#	Time	Action	Message	Duration / Fetch
52	13:41:27	SELECT * FROM class LEFT JOIN class_info ON class_id = class_info_id WHERE class_info_id IS NULL UNION...	4 row(s) returned	0.000 sec / 0.000 sec
53	13:42:06	INSERT INTO second_table VALUES (2, 'adam'), (3, 'chester')	Error Code: 1146 Table 'school_db.second_table' doesn't exist	0.000 sec
54	13:42:14	CREATE TABLE first_table (id INT, name VARCHAR(30))	0 row(s) affected	0.000 sec
55	13:42:17	CREATE TABLE second_table (id INT, name VARCHAR(30))	0 row(s) affected	0.015 sec
56	13:42:21	INSERT INTO first_table VALUES (1, 'abhi'), (2, 'adam')	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0	0.015 sec
57	13:42:24	INSERT INTO second_table VALUES (2, 'adam'), (3, 'chester')	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0	0.000 sec

11. Union Operations

```

99 • SELECT * FROM first_table
100
101 • SELECT * FROM second_table;
102
103 • SELECT name FROM first_table
104
105 • SELECT name FROM second_table;

```

Result Grid

id	name
1	abhi
2	adam
3	chester

Result 15

#	Time	Action	Message	Duration / Fetch
55	13:42:17	CREATE TABLE second_table (id INT, name VARCHAR(30))	0 row(s) affected	0.015 sec
56	13:42:21	INSERT INTO first_table VALUES (1, 'abhi'), (2, 'adam')	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0	0.015 sec
57	13:42:24	INSERT INTO second_table VALUES (2, 'adam'), (3, 'chester')	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0	0.000 sec
58	13:42:57	SELECT name FROM first_table UNION SELECT name FROM second_table	3 row(s) returned	0.000 sec / 0.000 sec
59	13:43:03	SELECT * FROM first_table UNION SELECT * FROM second_table	3 row(s) returned	0.000 sec / 0.000 sec
60	13:43:10	SELECT * FROM first_table UNION SELECT * FROM second_table	3 row(s) returned	0.000 sec / 0.000 sec

12. Union All Operations

```

109
110 • SELECT COUNT(*)
111 FROM (
112   SELECT * FROM first_table
113   UNION ALL
114   SELECT * FROM second_table
115 ) AS total_records;

```

Result Grid

COUNT(*)
4

Result 16

#	Time	Action	Message	Duration / Fetch
56	13:42:21	INSERT INTO first_table VALUES (1, 'abhi'), (2, 'adam')	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0	0.015 sec
57	13:42:24	INSERT INTO second_table VALUES (2, 'adam'), (3, 'chester')	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0	0.000 sec
58	13:42:57	SELECT name FROM first_table UNION SELECT name FROM second_table	3 row(s) returned	0.000 sec / 0.000 sec
59	13:43:03	SELECT * FROM first_table UNION SELECT * FROM second_table	3 row(s) returned	0.000 sec / 0.000 sec
60	13:43:10	SELECT * FROM first_table UNION SELECT * FROM second_table	3 row(s) returned	0.000 sec / 0.000 sec
61	13:44:10	SELECT COUNT(*) FROM (SELECT * FROM first_table UNION ALL SELECT * FROM second_table) AS total...	1 row(s) returned	0.000 sec / 0.000 sec

13. Intersect Operations

119 • `SELECT name FROM first_table`

Result Grid

id	name
2	adam

Result 23 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
64	13:44:52	<code>SELECT * FROM first_table INTERSECT SELECT * FROM second_table</code>	1 row(s) returned	0.000 sec / 0.000 sec
65	13:44:57	<code>SELECT * FROM first_table INTERSECT SELECT * FROM second_table</code>	1 row(s) returned	0.000 sec / 0.000 sec
66	13:45:03	<code>SELECT name FROM first_table INTERSECT SELECT name FROM second_table</code>	1 row(s) returned	0.000 sec / 0.000 sec
67	13:45:10	<code>SELECT name FROM first_table INTERSECT SELECT name FROM second_table</code>	1 row(s) returned	0.000 sec / 0.000 sec
68	13:45:13	<code>INTERSECT SELECT name FROM second_table</code>	Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds to your MySQL ser...	0.000 sec
69	13:45:36	<code>SELECT * FROM first_table INTERSECT SELECT * FROM second_table</code>	1 row(s) returned	0.000 sec / 0.000 sec

14. Minus / Except Operations

123 • `SELECT * FROM first_table`

124 `EXCEPT`

125 `SELECT * FROM second_table;`

126

127 • `SELECT name FROM first_table`

128 `EXCEPT`

129 `SELECT name FROM second_table;`

Result Grid

name
abhi

Result 26 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
71	13:46:37	<code>USE school_db</code>	0 row(s) affected	0.000 sec
72	13:46:37	<code>CREATE TABLE class (id INT, name VARCHAR(30))</code>	Error Code: 1050. Table 'class' already exists	0.000 sec
73	13:46:41	<code>SELECT name FROM first_table SELECT * FROM first_table EXCEPT SELECT * FROM second_table</code>	Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds to your MySQL ser...	0.000 sec
74	13:49:44	<code>SELECT name FROM first_table INTERSECT SELECT name FROM second_table</code>	1 row(s) returned	0.000 sec / 0.000 sec
75	13:49:49	<code>SELECT name FROM first_table INTERSECT SELECT name FROM second_table</code>	1 row(s) returned	0.000 sec / 0.000 sec
76	13:50:07	<code>SELECT name FROM first_table EXCEPT SELECT name FROM second_table</code>	1 row(s) returned	0.000 sec / 0.000 sec

15. Set Difference Alternative

129 `SELECT name FROM second_table;`

130 • `SELECT f.*`

131 `FROM first_table f`

132 `LEFT JOIN second_table s ON f.id = s.id`

133 `WHERE s.id IS NULL;`

134

135 • `SELECT f.name`

136 `FROM first_table f`

137 `LEFT JOIN second_table s ON f.name = s.name`

138 `WHERE s.name IS NULL;`

Result Grid

name
abhi

Result 30 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
75	13:49:49	<code>SELECT name FROM first_table INTERSECT SELECT name FROM second_table</code>	1 row(s) returned	0.000 sec / 0.000 sec
76	13:50:07	<code>SELECT name FROM first_table EXCEPT SELECT name FROM second_table</code>	1 row(s) returned	0.000 sec / 0.000 sec
77	13:50:53	<code>SELECT f.name FROM first_table f LEFT JOIN second_table s ON f.name = s.name WHERE s.name IS NULL ...</code>	1 row(s) returned	0.000 sec / 0.000 sec
78	13:50:58	<code>SELECT f.* FROM first_table f LEFT JOIN second_table s ON f.id = s.id WHERE s.id IS NULL LIMIT 0, 1000</code>	1 row(s) returned	0.000 sec / 0.000 sec
79	13:51:00	<code>SELECT f.* FROM first_table f LEFT JOIN second_table s ON f.id = s.id WHERE s.id IS NULL LIMIT 0, 1000</code>	1 row(s) returned	0.000 sec / 0.000 sec
80	13:51:05	<code>SELECT f.name FROM first_table f LEFT JOIN second_table s ON f.name = s.name WHERE s.name IS NULL ...</code>	1 row(s) returned	0.000 sec / 0.000 sec

16. Advanced Queries with Aliases and Case Expressions

```

144 • SELECT c.id,
145       c.name,
146       CASE
147       WHEN ci.address IS NULL THEN 'Address Missing'
148       ELSE 'Address Available'
149       END AS Status
150 FROM class c
151 LEFT JOIN class_info ci
152 ON c.id = ci.id;

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

	id	name	Status
▶	1	abhi	Address Available
	2	adam	Address Available
	3	alex	Address Available
	4	anu	Address Missing
	5	ashish	Address Missing

Result Grid
Form Editor

Result 31 x Read Only

Output
Action Output

#	Time	Action	Message	Duration / Fetch
✓	76 13:50:07	SELECT name FROM first_table EXCEPT SELECT name FROM second_table	1 row(s) returned	0.000 sec / 0.000 sec
✓	77 13:50:53	SELECT f.name FROM first_table f LEFT JOIN second_table s ON f.name = s.name WHERE s.name IS NULL ...	1 row(s) returned	0.000 sec / 0.000 sec
✓	78 13:50:58	SELECT f.* FROM first_table f LEFT JOIN second_table s ON f.id = s.id WHERE s.id IS NULL LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec
✓	79 13:51:00	SELECT f.* FROM first_table f LEFT JOIN second_table s ON f.id = s.id WHERE s.id IS NULL LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec
✓	80 13:51:05	SELECT f.name FROM first_table f LEFT JOIN second_table s ON f.name = s.name WHERE s.name IS NULL ...	1 row(s) returned	0.000 sec / 0.000 sec
✓	81 13:51:58	SELECT c.id, c.name, CASE WHEN ci.address IS NULL THEN 'Address Missing' ELSE 'Address Available' EN...	5 row(s) returned	0.000 sec / 0.000 sec